

Meeting 2 — Exercises

Total Differentials, Exact Forms, and Line Integrals

Mathematics for Chemists

The exercises below follow the structure of the lecture. Exercises 1–3 concern total differentials and the chain rule; Exercises 4–8 concern exactness and finding potentials; Exercises 9–13 are line integrals; Exercises 14–16 are chemistry applications.

Total Differentials and the Chain Rule

Zadanie 1. Compute the total differential df of each of the following functions.

- (a) $f(x, y) = x^3 - 2xy + y^2$
- (b) $f(x, y) = e^{xy} \sin y$
- (c) $f(x, y, z) = xyz$
- (d) $f(x, y, z) = \ln(x^2 + y^2 + z^2)$
- (e) $f(x, y) = \arctan(y/x)$ (for $x > 0$)

Odpowiedź: (a) $df = (3x^2 - 2y) dx + (-2x + 2y) dy$; (b) $df = ye^{xy} \sin y dx + (xe^{xy} \sin y + e^{xy} \cos y) dy$; (c) $df = yz dx + xz dy + xy dz$; (d) $df = \frac{2x dx + 2y dy + 2z dz}{x^2 + y^2 + z^2}$; (e) $df = \frac{-y dx + x dy}{x^2 + y^2}$.

Zadanie 2. For the function $f(x, y) = x^2 + y^2$, verify numerically at the point $(1, 1)$ that df gives a good linear approximation to Δf for the displacement $(\Delta x, \Delta y) = (0.1, 0.2)$. How large is the error $|\Delta f - df|$ relative to $\|(\Delta x, \Delta y)\|$?

Odpowiedź: $\Delta f = (1.1)^2 + (1.2)^2 - 2 = 0.65$; $df = 2(0.1) + 2(0.2) = 0.6$; error = 0.05. $\|(\Delta x, \Delta y)\| = \sqrt{0.05} \approx 0.224$, so error/ $\|(\Delta x, \Delta y)\| \approx 0.22 \approx \|(\Delta x, \Delta y)\|$. The error is of order $\|(\Delta x, \Delta y)\|^2$, confirming the linear approximation.

Zadanie 3. Each of the following describes a physical quantity depending on two or more variables that are themselves functions of time. Write the total derivative with respect to t using the chain rule.

- (a) The internal energy of a van der Waals gas is $U(T, V) = nC_V T - an^2/V$. A gas sample undergoes a process where $T(t) = T_0 e^{-t}$ and $V(t) = V_0(1 + t)$. Find dU/dt .
- (b) The equilibrium constant satisfies $\ln K = -\Delta H^\circ/(RT) + \Delta S^\circ/R$ (constant $\Delta H^\circ, \Delta S^\circ$). Compute $d(\ln K)/dT$ and name the resulting equation.
- (c) The refractive index $n(c, T)$ of a solution changes with concentration c and temperature T . In a refractometry experiment the concentration changes with temperature at rate $dc/dT = \alpha$. Write the total observed dn/dT .

Odpowiedź: (a) $dU/dt = nC_V(-T_0 e^{-t}) + \frac{an^2}{V_0^2(1+t)^2} \cdot V_0 = (an^2/V(t)^2) \cdot V_0 - nC_V T_0 e^{-t}$.
(b) $d(\ln K)/dT = \Delta H^\circ/RT^2$ — the **van't Hoff equation**. (c) $dn/dT = (\partial n/\partial T)_c +$

$$(\partial n / \partial c)_T \cdot \alpha.$$

Exactness and Finding Potentials

Zadanie 4. Test whether each expression is exact. For those that are, find the potential function f .

- (a) $y \, dx + x \, dy$
- (b) $y \, dx - x \, dy$
- (c) $3x^2y \, dx + x^3 \, dy$
- (d) $(2xy - \sin x) \, dx + (x^2 + e^y) \, dy$
- (e) $\frac{x \, dx + y \, dy}{\sqrt{x^2 + y^2}}$

Wskazówka: For (e), write the numerator as $\frac{1}{2}d(x^2 + y^2)$ and recall $d(\sqrt{u}) = du/(2\sqrt{u})$.

Odpowiedź: (a) Exact; $f = xy$. (b) Not exact: $\partial_y P = 1 \neq -1 = \partial_x Q$. (c) Exact: $\partial_y(3x^2y) = 3x^2 = \partial_x(x^3)$; $f = x^3y$. (d) Exact: $\partial_y(2xy - \sin x) = 2x = \partial_x(x^2 + e^y)$; $f = x^2y + \cos x + e^y$. (e) Exact; $f = \sqrt{x^2 + y^2}$.

Zadanie 5. The expression $(2xy + y^2) \, dx + (x^2 + 2xy + 1) \, dy$ is exact. Find the potential $f(x, y)$ such that $f(0, 0) = 0$.

Odpowiedź: $f = x^2y + xy^2 + y$. Check: $\partial_x f = 2xy + y^2$, $\partial_y f = x^2 + 2xy + 1$.

Zadanie 6. Test whether $(yz + z) \, dx + xz \, dy + (xy + x) \, dz$ is exact in three variables (all three conditions $\partial_y P = \partial_x Q$, $\partial_z P = \partial_x R$, $\partial_z Q = \partial_y R$ must hold). If so, find the potential.

Odpowiedź: All three conditions hold. $f = xyz + xz$. Check: $\partial_x f = yz + z$, $\partial_y f = xz$, $\partial_z f = xy + x$.

Zadanie 7. The expression $(axy^2 + by) \, dx + (x^2y + bx + cy^2) \, dy$ is exact for certain values of the constants $a, b, c \in \mathbb{R}$. Find all such values and, for those values, find the potential.

Wskazówka: Write $P = axy^2 + by$, $Q = x^2y + bx + cy^2$. The exactness condition $\partial_y P = \partial_x Q$ must hold for all (x, y) .

Odpowiedź: $\partial_y P = 2axy + b$; $\partial_x Q = 2xy + b$. Matching coefficients: $2a = 2 \Rightarrow a = 1$; b arbitrary; c arbitrary. So $a = 1$, $b, c \in \mathbb{R}$ arbitrary. Potential: $f = \frac{1}{2}x^2y^2 + bxy + \frac{c}{3}y^3$.

Zadanie 8. (Chemistry) The generalised Gibbs relation in a multicomponent system is

$$dG = -S \, dT + V \, dp + \sum_i \mu_i \, dn_i,$$

where μ_i is the chemical potential of species i and n_i its amount. Write the Maxwell relation obtained by comparing $\partial^2 G / \partial T \partial n_i$ and $\partial^2 G / \partial n_i \partial T$. What is the physical content of the resulting identity?

Odpowiedź: From dG : $(\partial G/\partial T)_{p,n} = -S$ and $(\partial G/\partial n_i)_{T,p,n_j} = \mu_i$. Schwarz gives: $(\partial \mu_i/\partial T)_{p,n} = -(\partial S/\partial n_i)_{T,p,n}$. Physical content: the rate at which the chemical potential of species i changes with temperature equals (minus) the partial molar entropy — the entropy contributed per mole of species i added at fixed T, p .

Line Integrals

Zadanie 9. Compute $\int_{\gamma} y \, dx + x \, dy$ from $(0, 0)$ to $(1, 1)$ along:

- (a) the straight line $y = x$;
- (b) the parabola $y = x^2$;
- (c) the path going first along the x -axis to $(1, 0)$, then vertically to $(1, 1)$.

Explain why all three answers are the same.

Odpowiedź: Check exactness first: $\partial_y(y) = 1 = \partial_x(x)$, so $y \, dx + x \, dy = d(xy)$. By the fundamental theorem: $\int_{\gamma} d(xy) = xy|_{(0,0)}^{(1,1)} = 1 - 0 = 1$ along *any* path. All three answers equal 1.

Zadanie 10. Compute $\int_{\gamma} (x^2 + y) \, dx + (y^2 + x) \, dy$ along:

- (a) the straight line from $(0, 0)$ to $(1, 1)$;
- (b) the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$;
- (c) the path: first horizontally to $(1, 0)$, then vertically to $(1, 1)$.

Test the exactness condition first and predict whether the answers should agree.

Wskazówka: $P = x^2 + y$, $Q = y^2 + x$: $\partial_y P = 1 = \partial_x Q$. Exact. Find f .

Odpowiedź: Exactness: $\partial_y P = 1 = \partial_x Q$, so the form is exact. Potential: $f = x^3/3 + xy + y^3/3$. All three integrals equal $f(1, 1) - f(0, 0) = 1/3 + 1 + 1/3 = 5/3$.

Zadanie 11. Compute $\int_{\gamma} x \, dy - y \, dx$ along:

- (a) the upper semicircle from $(-1, 0)$ to $(1, 0)$;
- (b) the lower semicircle from $(-1, 0)$ to $(1, 0)$.

Do the results agree? What does the difference represent geometrically?

Wskazówka: Parametrise the upper semicircle as $x = \cos t$, $y = \sin t$, $t \in [0, \pi]$. First test: is $x \, dy - y \, dx$ exact?

Odpowiedź: Not exact: $\partial_y(-y) = -1 \neq 1 = \partial_x(x)$.
Upper semicircle: $\int_0^{\pi} (\cos t \cos t - (-\sin t)(-\sin t)) \, dt = \int_0^{\pi} (\cos^2 t - \sin^2 t) \, dt = \int_0^{\pi} \cos 2t \, dt = 0$. Wait — let us redo carefully. $x \, dy - y \, dx = \cos t(\cos t \, dt) - \sin t(-\sin t \, dt) = (\cos^2 t + \sin^2 t) \, dt = dt$. So integral $= \pi$.
Lower semicircle ($y = -\sin t$, $t \in [0, \pi]$): $x \, dy - y \, dx = \cos t(-\cos t \, dt) -$

$(-\sin t)(-\sin t \, dt) = -dt$. Integral $= -\pi$.

The results differ: $\pi \neq -\pi$. The form is non-exact. The difference $\pi - (-\pi) = 2\pi$ equals twice the area of the unit semicircle — in fact $x \, dy - y \, dx = 2 \, dA$ in polar coordinates, and the closed-loop integral equals twice the enclosed area (this is Green's theorem, which we will prove later).

Zadanie 12. The work done by a force field $\mathbf{F} = (xy, x^2)$ on a particle moving from $(0, 0)$ to $(2, 1)$:

- (a) along the straight line $y = x/2$;
- (b) along the x -axis to $(2, 0)$, then vertically to $(2, 1)$.

Show the results differ, confirming $\mathbf{F}$ is not conservative. Verify this using the exactness condition.

Odpowiedź: Exactness check: $P = xy$, $Q = x^2$; $\partial_y P = x \neq 2x = \partial_x Q$ (unless $x = 0$). Not exact.

(a) Line $y = t/2$, $x = t$, $t \in [0, 2]$: $\int_0^2 (t \cdot \frac{t}{2} \cdot 1 + t^2 \cdot \frac{1}{2}) \, dt = \int_0^2 t^2 \, dt = 8/3$.

(b) Along x -axis ($y = 0$): $P = 0$, so work $= 0$. Vertical segment ($x = 2$, $y \in [0, 1]$): $\int_0^1 x^2 \, dy = \int_0^1 4 \, dy = 4$. Total: $0 + 4 = 4 \neq 8/3$.

Zadanie 13. (Preview of topology.) Compute $\int_{\gamma} \frac{-y}{x^2 + y^2} \, dx + \frac{x}{x^2 + y^2} \, dy$ along:

- (a) the unit circle $x = \cos t$, $y = \sin t$, $t \in [0, 2\pi]$;
- (b) the circle of radius 2 traversed twice counterclockwise;
- (c) a small circle around $(3, 0)$ not enclosing the origin.

Verify $\partial_y P = \partial_x Q$ where defined. Explain why the form is nevertheless non-exact.

Odpowiedź: Check: $\partial_y P = \partial_x Q = (y^2 - x^2)/(x^2 + y^2)^2$. Equal.

(a) Substitute: integrand $= dt$. Integral $= 2\pi$.

(b) Same calculation, traversed twice: 4π .

(c) The origin is not enclosed. The form is locally exact near $(3, 0)$ (since $(0, 0)$ is not in a small disk around $(3, 0)$), so the integral around a small loop there is 0.

The form is closed ($\partial_y P = \partial_x Q$) but *not* exact on $\mathbb{R}^2 \setminus \{(0, 0)\}$: the domain has a hole, and the exactness theorem requires a simply connected domain. The integral equals 2π times the winding number of γ around the origin.

Chemistry Applications

Zadanie 14. (Isothermal vs. irreversible expansion.) A mole of ideal gas expands from V_1 to $V_2 = 2V_1$ at temperature T .

- (a) Compute the work $W = \int_{V_1}^{V_2} p \, dV$ for a reversible isothermal expansion ($p = RT/V$).
- (b) Compute W for an irreversible expansion against constant external pressure $p_{\text{ext}} = RT/V_2$.
- (c) Which is larger? What does this say about the efficiency of reversible processes?

Odpowiedź: (a) $W = RT \ln 2$. (b) $W = p_{\text{ext}}(V_2 - V_1) = \frac{RT}{2V_1} \cdot V_1 = \frac{RT}{2}$. (c) $RT \ln 2 \approx 0.693 RT > 0.5 RT$. The reversible process extracts more work. This is a general principle: reversible work is the maximum obtainable work.

Zadanie 15. (Thermodynamic cycle.) An ideal gas ($n = 1$ mole) undergoes the cycle: (i) isothermal expansion at T_1 from V_1 to $2V_1$; (ii) isochoric cooling from T_1 to T_2 ; (iii) isothermal compression at T_2 from $2V_1$ back to V_1 ; (iv) isochoric heating from T_2 back to T_1 .

- (a) Compute the total work $W = \oint p dV$ done in the cycle.
- (b) Why is $\oint dU = 0$ regardless of the path?
- (c) Express the net work in terms of T_1 , T_2 , and R , and verify its sign makes physical sense.

Odpowiedź: (a) Steps (ii) and (iv) are isochoric ($dV = 0$), contributing no work. Steps (i) and (iii) give $W_1 = RT_1 \ln 2$ and $W_3 = -RT_2 \ln 2$. Total: $W = (T_1 - T_2)R \ln 2$.
 (b) dU is exact; U is a state function. The integral of an exact form around any closed loop is zero.
 (c) If $T_1 > T_2$, then $W > 0$: the system does net work on the surroundings. This is a Carnot-like cycle and the net work equals the heat absorbed times the thermodynamic efficiency.

Zadanie 16. (Conservative fields in spectroscopy.) The Beer–Lambert law gives the absorbance $A = \varepsilon c \ell$, where ε is the molar absorptivity, c the concentration, and ℓ the path length. In a spatially inhomogeneous sample, $\varepsilon = \varepsilon(x, y)$ and the total absorbance along a beam path γ is

$$A = \int_{\gamma} \varepsilon(x, y) ds,$$

where $ds = \sqrt{dx^2 + dy^2}$ is the arc length element.

- (a) For a homogeneous sample ($\varepsilon = \varepsilon_0$ constant), show that A depends only on the total path length, not the shape of γ .
- (b) For an inhomogeneous sample where $\varepsilon = e^{-x^2 - y^2}$ (a Gaussian concentration profile), compute A along the straight line from $(0, 0)$ to $(1, 0)$ and along the parabola $y = x(1 - x)$ from $(0, 0)$ to $(1, 0)$. Do they agree? Should they?

Wskazówka: For (b): this is *not* the same as a line integral of the form $P dx + Q dy$. The arc length element mixes dx and dy . Along the x -axis: $ds = dx$. Along the parabola: $ds = \sqrt{1 + (1 - 2x)^2} dx$.

Odpowiedź: (a) $A = \varepsilon_0 \int_{\gamma} ds = \varepsilon_0 \cdot \ell$, where ℓ is the geometric length of γ . This depends on the shape (length) of the path, not the endpoints — even for a constant ε .
 (b) Along x -axis: $A_1 = \int_0^1 e^{-x^2} dx \approx 0.747$. Along parabola: $A_2 = \int_0^1 e^{-x^2 - x^2(1-x)^2} \sqrt{1 + (1 - 2x)^2} dx$. The integrand differs and $A_2 \neq A_1$. They should *not* agree: the absorbance integral is an arc-length integral weighted by the local absorptivity, which depends on which points the beam passes through — a genuinely path-dependent quantity, like work.